

SAN FRANCISCO INTL, CA, USA

WMO: 724940

Lat: 37.620N

Lon: 122.365W

Elev: 2

StdP: 101.30

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23234

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.6	(c) 5.7	(d) -3.2	(e) 2.9	(f) 11.2	(g) -1.2	(h) 3.4	(i) 10.8	(j) 14.0	(k) 13.5	(l) 11.8	(m) 12.9	(n) 1.6	(o) 150	(p) 0.583

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	8.9	28.3	17.1	25.6	16.7	23.6	16.4	18.8	25.2	17.9	23.6	17.2	22.3	5.6	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.4	11.7	20.5	15.7	11.1	19.7	15.0	10.7	19.1	53.1	25.2	50.3	23.6	48.1	22.2	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.8	11.4	10.4	2.4	33.9	1.2	2.0	1.5	35.3	0.8	36.5	0.2	37.6	-0.7	39.0
			0.7	20.8	1.4	1.3	-0.3	21.7	-1.0	22.5	-1.8	23.2	-2.8	24.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.6	10.7	11.7	12.9	13.8	15.1	16.6	17.4	17.9	18.2	16.8	13.4	10.8		
	DBStd	3.49	2.10	2.22	2.31	2.41	2.36	2.35	1.83	1.91	2.59	2.53	2.34	2.28		
	HDD10.0	47	16	8	2	1	0	0	0	0	0	0	3	17		
	HDD18.3	1448	236	187	169	139	106	62	41	31	32	63	148	234		
	CDD10.0	1738	39	54	92	114	159	199	229	245	247	210	106	42		
	CDD18.3	96	0	0	1	3	7	12	12	18	29	15	1	0		
	CDH23.3	525	0	0	6	26	50	86	54	69	154	78	3	0		
	CDH26.7	148	0	0	0	6	14	29	12	19	49	19	0	0		
(14) Wind	WSAvg	4.6	2.9	3.7	4.5	5.5	6.1	6.1	5.7	5.4	4.9	4.0	3.1	3.1		
(15) Precipitation	PrecAvg	556	106	103	75	37	19	5	1	2	3	28	59	118		
	PrecMax	966	263	362	220	128	84	41	2	22	12	95	182	300		
	PrecMin	206	7	0	4	2	0	0	0	0	0	0	2	10		
	PrecStd	160	78	75	61	30	20	9	0	4	3	27	45	81		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.4	21.1	23.7	26.7	28.6	30.4	28.0	29.5	31.6	29.2	22.8	18.0		
		MCWB	11.7	13.8	15.0	16.0	16.5	17.5	18.4	18.2	18.0	16.1	15.2	14.1		
	2%	DB	16.3	18.1	20.2	22.3	23.7	25.4	24.4	24.8	27.5	25.3	20.0	16.4		
		MCWB	12.3	13.3	13.7	14.4	15.3	16.5	17.3	17.6	17.4	15.3	14.1	13.5		
	5%	DB	15.0	16.4	17.9	19.2	20.7	22.4	22.4	22.8	24.4	22.8	18.4	15.4		
		MCWB	11.9	12.8	13.1	13.4	14.8	15.9	16.6	17.0	17.0	15.1	13.7	12.9		
	10%	DB	14.0	15.1	16.4	17.4	18.7	20.6	21.0	21.4	22.3	20.8	17.1	14.5		
		MCWB	11.5	12.1	12.6	12.7	13.7	15.1	15.9	16.3	16.6	15.0	13.5	12.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.9	15.7	16.4	17.1	18.5	19.0	19.1	19.4	20.2	18.6	16.9	16.0		
		MCDB	16.0	17.8	21.1	23.9	25.1	27.7	26.4	25.8	26.3	24.0	20.2	17.0		
	2%	WB	13.6	14.5	15.0	15.4	16.6	17.3	17.7	18.1	18.8	17.4	15.8	14.9		
		MCDB	14.8	16.8	18.5	20.3	21.9	24.0	23.5	23.5	24.9	22.2	18.4	15.9		
	5%	WB	12.7	13.5	14.0	14.3	15.2	16.3	16.9	17.3	17.8	16.6	15.0	13.8		
		MCDB	14.1	15.7	16.7	18.2	19.6	21.9	21.9	22.2	23.3	21.1	17.4	14.8		
	10%	WB	11.9	12.7	13.2	13.2	14.2	15.3	16.1	16.6	16.8	15.6	14.1	12.6		
		MCDB	13.3	14.6	15.7	16.7	18.1	20.1	20.6	21.2	21.7	19.7	16.3	13.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.9	6.5	7.3	7.7	7.6	8.4	8.1	8.1	8.9	8.6	6.9	5.8		
		MCDBR	7.4	8.7	10.1	11.4	11.8	12.4	10.4	10.4	13.3	12.3	8.9	6.4		
		MCWBR	4.5	5.1	5.7	5.9	5.7	5.8	4.9	4.7	6.2	5.9	4.9	4.3		
	5% WB	MCDBR	5.3	7.3	8.6	10.2	10.5	11.6	10.1	10.1	12.0	10.5	7.3	4.6		
		MCWBR	3.9	4.6	5.3	5.7	5.4	5.6	4.8	4.7	5.8	5.5	4.6	3.9		
(40) Clear-Sky Solar Irradiance	taub		0.324	0.332	0.341	0.358	0.365	0.359	0.367	0.345	0.348	0.338	0.322	0.324		
	taud		2.562	2.537	2.504	2.421	2.426	2.449	2.432	2.494	2.470	2.540	2.568	2.566		
	Ebn at Noon		866	903	925	922	915	918	907	923	904	882	860	839		
	Edh at Noon		76	87	99	114	115	112	114	104	101	86	74	71		
(44) All-Sky Solar Radiation	RadAvg		2.41	3.33	4.71	6.16	7.00	7.60	7.08	6.30	5.41	4.03	2.82	2.10		
	RadStd		0.32	0.40	0.41	0.47	0.47	0.33	0.26	0.31	0.22	0.26	0.18	0.30		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	-1.12	N/A	N/A	N/A	N/A	N/A	+136	+22
(47) Regional (44 neighbors)		N/A	N/A	-1.19	N/A	N/A	N/A	N/A	-110	+135	N/A

Nomenclature: See separate page